

CONTINUOUS AUTHENTICATION

Watchdog for the untrusted workspace.

Your desktop verifies who is sitting in front of it — every few seconds, on-device, with no network and no cloud. When verification fails, the screen protects itself.

THREE WAYS A WORKSTATION IS EXPOSED

01 ABSENCE

You step away from the desk.

Open windows minimise, and keyboard, mouse and USB input stop responding. Everything goes back to normal the instant your face returns — no password, no interruption.

02 OBSERVER

Someone behind you looks at your screen.

If an unknown face keeps looking for `n_seconds`, the screen shows a warning and the event is recorded. A quick glance never counts.

03 ENVIRONMENT

The machine leaves its trusted area.

A small wireless node travels with the desk and marks where the machine belongs. If the node goes out of range and no approved user is present, the machine locks itself.

Hsu-Ming Chien & Rivaldo Fonkou · Professor Timothy Wong ·
Sponsor Darryl Wood

Computer System Technician · Humber Polytechnic · 2026



Sensors don't decide. Sentinel does.

Every layer has one job: sensors report facts, a classifier names them, and one engine holds the whole policy.

LAYER 1 · SIGNALS

MonitoringService

Watches the camera, finds faces and checks them against the enrolled user. Reports `authorized_user`.

TrustedNodeService

Listens for the trusted anchor and checks its identity and signal strength. Reports `node_signal`.

Senses - Never judges

LAYER 2 · EVENT SERVICE

NONE

ON_OBSERVER

ON_NODE_LOSS

ON_USER_ABSENT

ON_UNSECURE_ENV

Classifies - Never judges

A label, not a value — and several can be true at once.

LAYER 3 · SENTINEL

`decide(state, events) → (state, commands)`

One pure function holds the priority order, the single user-set timer, and every escalation path.

The only decision maker

LAYER 4 · ACTIONS

Notify

Popup and audit entry.

Contain

Overlay, minimise, block input.

Lock

Hands the machine to the system lock screen.

Executes - Never judges

STATE MODEL

BASELINE

SCANNING

Verified user present. Nothing is hidden.

← ON USER ABSENT PRIVACYMODE

App-level containment. The camera clears it the moment you return.

← ON UNSECURE ENV LOCKDOWN

The real OS lock. Windows credentials only.

One glance tells you the state of the room.

A calm dashboard while everything is verified — and an unambiguous change of state the moment it is not.

● SECURE · AUTHORIZED USER VERIFIED

Home dashboard

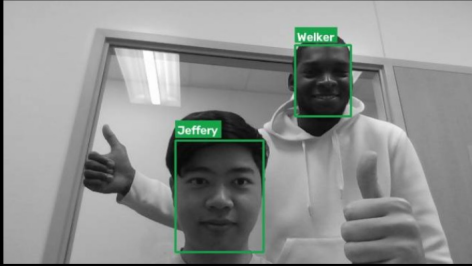
SentinelGuard

Home

Home

Security

Logs



System Status

Identity Verification

Continuous user authentication

VERIFIED

Privacy Monitoring

Observer and visibility detection

CLEAR

Environment Trust

Trusted device proximity

VERIFIED

Proximity Telemetry

Weak

Strong

CONNECTED

STRONG

Diagnostics

RSSI (raw)

-62 dBm

RSSI (EWMMA)

-60 dBm

Device

04:E9:F4:8A:C8:12

Last Seen

Live

Recent Activity

TITLE	STATUS	DETAIL	TIMESTAMP
Biometric profile	Success	New biometric profile registered Welker	2026-08-06 03:23:58 PM
Absence timeout	Success	Absence timeout updated 10s → 5s	2026-08-02 03:25:00 PM
Auto-connect	Success	Trusted Node auto-connect enabled	2026-08-02 03:23:46 PM

● PRIVACYPAGE ENGAGED



● TRUSTED NODE ANCHOR



Hsu-Ming Chien & Rivaldo Fonkou · Professor Timothy Wong ·
Sponsor Darryl Wood

Computer System Technician · Humber Polytechnic · 2026

